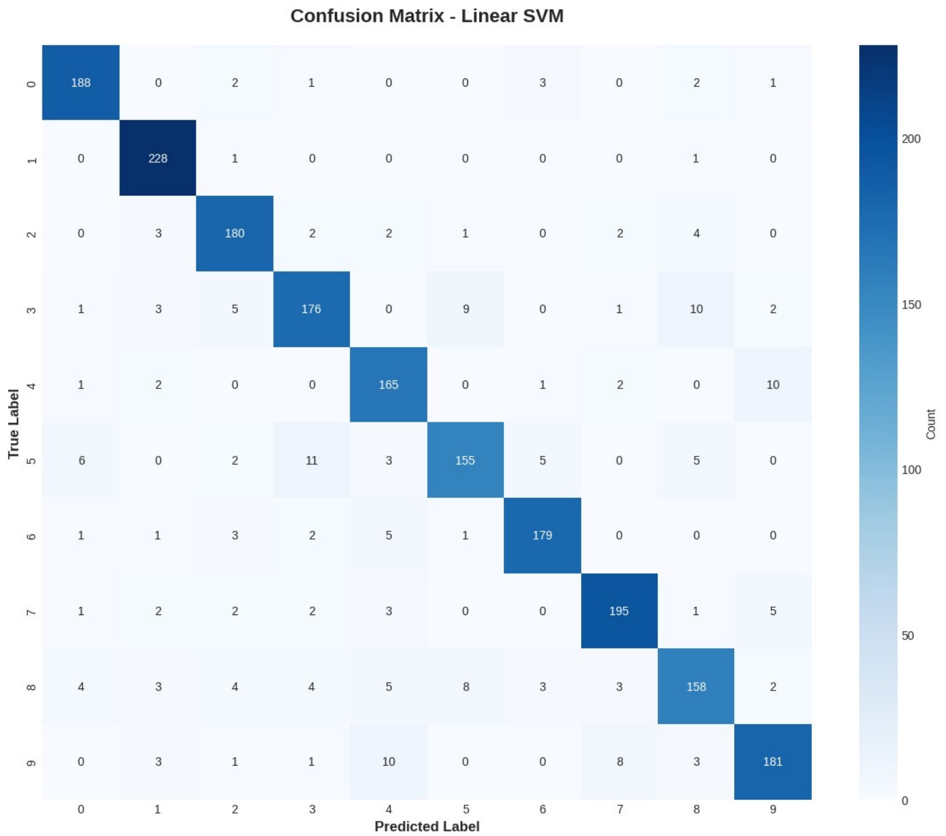
Q. *How can a Support Vector Machine (SVM) be used to classify MNIST handwritten digits and evaluated using accuracy, confusion matrix, and classification report?*

Ans:

1. Linear SVM – Predictions:

Linear SVM Accuracy: 90.25%
Confusion Matrix:

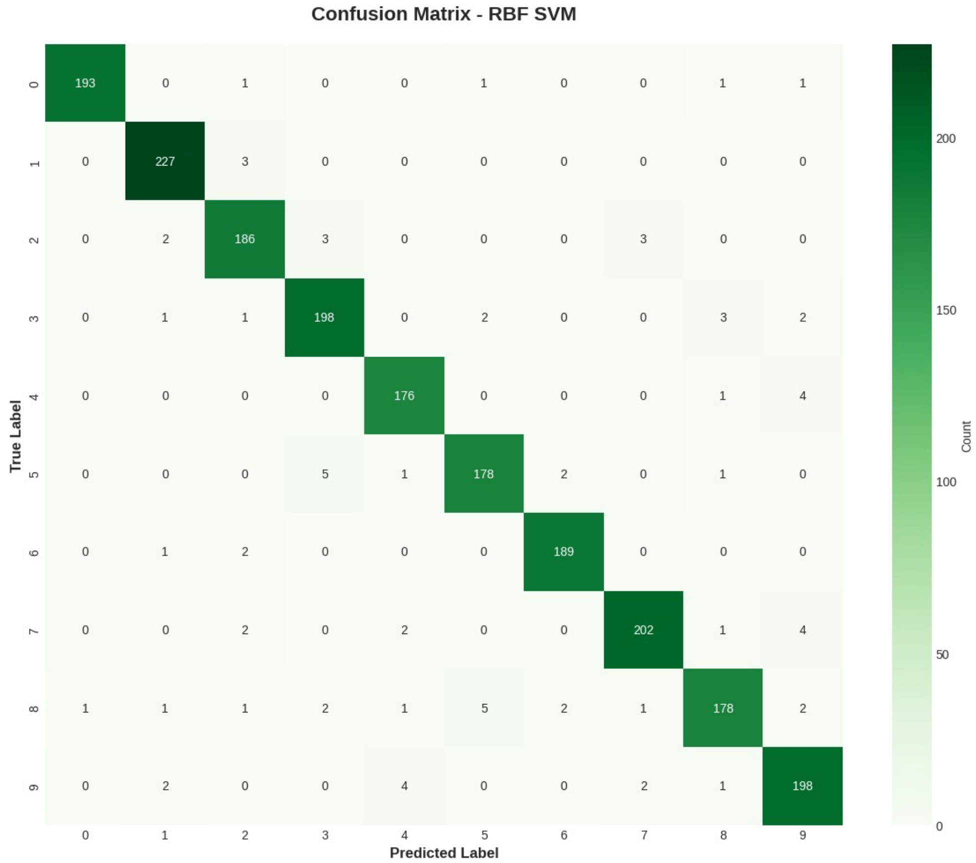


Linear SVM Classification Report:

	precision	recall	f1-score	support
0	0.93	0.95	0.94	197
1	0.93	0.99	0.96	230
2	0.90	0.93	0.91	194
3	0.88	0.85	0.87	207
4	0.85	0.91	0.88	181
5	0.89	0.83	0.86	187
6	0.94	0.93	0.93	192
7	0.92	0.92	0.92	211
8	0.86	0.81	0.84	194
9	0.90	0.87	0.89	207
accuracy			0.90	2000
macro avg	0.90	0.90	0.90	2000
weighted avg	0.90	0.90	0.90	2000

2. Non-Linear SVM Predictions:

SVM Accuracy: 96.25%
Confusion Matrix (RBF SVM):



RBF SVM Classification Report:

	precision	recall	f1-score	support
0	0.99	0.98	0.99	197
1	0.97	0.99	0.98	230
2	0.95	0.96	0.95	194
3	0.95	0.96	0.95	207
4	0.96	0.97	0.96	181
5	0.96	0.95	0.95	187
6	0.98	0.98	0.98	192
7	0.97	0.96	0.96	211
8	0.96	0.92	0.94	194
9	0.94	0.96	0.95	207
accuracy			0.96	2000
macro avg	0.96	0.96	0.96	2000
weighted avg	0.96	0.96	0.96	2000